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Splitting algorithms for the high-order compact finite-difference schemes in wave-equation modeling

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ABSTRACT

We have developed efficient splitting algorithms for high-order compact finite-difference methods to approximate second-order space derivatives. In general, the methods' high-order compact finite-difference schemes require the inversion of a multidiagonal matrix that is commonly less efficient. To solve this problem, we used ideas from splitting algorithms in one-way wave-equation migration that work by decomposing the multidiagonal matrix into a series of tridiagonal matrices and then subsequently solving the tridiagonal matrices. This approach results in more efficient algorithms with little loss of accuracy. The splitting algorithms can be implemented in three different ways. Our computational complexity analysis verifies that our methods can reduce the calculation burden from exponential to linear growth. Numerical experiments demonstrate the correctness and effectiveness of our algorithms.

INTRODUCTION

Wave-equation forward modeling is one of the most important techniques used in seismic exploration. The well-known reverse time migration and full-waveform inversion methods rely on a forward modeling engine that can quickly and accurately simulate the wavefield. The finite-difference method (FDM) is a simple and efficient method that approximates the derivatives at one point of a function using its neighborhood's values (Kelly et al., 1976; Virieux, 1984, 1986; Levander, 1988; Lele, 1992; Tam and Webb, 1993; Moczo et al., 2000; Etgen and O'Brien, 2007; Bansal and Sen, 2008; Hestholm, 2009; Liu, 2013). In general, the FDM can be classified into two categories: explicit FD (EFD) and compact or implicit FD (CFD). EFD is a direct, simple, and widely used method. It commonly approximates the derivatives in the time direction with a second-order scheme while approximating the derivatives in the space direction with a high-order scheme to suppress the numerical grid dispersion (Dablain, 1986). The coefficients used in the high-order scheme by EFD are obtained by expressing the function around the zero wavenumber using Taylor's theorem (Dablain, 1986; Kindelan, 1990). Thus, it maintains relatively high accuracy in low-wavenumber zones, although the resolved error is significant in the high-wavenumber zones. As the size of the stencils increase, the improvement in accuracy is less obvious and the so-called "saturation effect" occurs (Kosloff et al., 2010). To overcome these problems, one can introduce optimized FDMs in which the optimized coefficients are used in the stencils by matching their spectral response with the exact one. This method improves the accuracy of the propagating waves with shorter wavelengths than conventional EFD (Robertsson et al., 1994; Ashcroft and Zhang, 2003; Shan, 2008; Zhou and Zhang, 2011; Chen, 2012).

However, the CFD approximates the spatial derivative operator using a rational expansion that converges faster than a power series. Lele (1992) proposes a pentadiagonal compact scheme for estimating the first and second derivatives. Similar to the optimized EFD methods, the optimized CFD methods (Kim and Lee, 1996; Zhou and Zhang, 2011; Liu, 2013) show features to the fourth order of accuracy. A spectral-like resolution can be achieved by optimizing the coefficients through fitting the derived wavenumber to the true one. Compared with EFD, CFD has a similar accuracy with smaller stencils, but a multidiagonal matrix (band matrix) needs to be inverted. Direct inversion of the band matrix would be less efficient, and it has strictly limited the application of the CFD methods. Prefactored compact schemes (Hixon, 1998, 2000; Ashcroft and Zhang, 2003; Zhou and Zhang, 2011) are some of the solutions used to improve the efficiency of the CFD methods. This method decom-

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poses the implicit matrix into independent upper and lower matrices that are easier to invert, and thus, the computational effort is reduced. Kosloff et al. (2008, 2010) implements CFD to acoustic- and elastic-wave simulations in a recursive way in which only tridiagonal matrices need to be inverted.

In addition to the spatial implicit algorithms, the implicit time-stepping methods are also unconditionally stable and have relatively large time steps (Kim and Lim, 2007; Chu and Stoffa, 2012). However, compared with implicit space methods, the direct implementation of the implicit time methods results in a larger matrix to invert. The alternating direction iteration (ADI) methods, which transform the original problem into a series of local 1D variants, can significantly reduce computations (Zhang et al., 2007; Nunez et al., 2013).

In this paper, we propose splitting algorithms for the high-order CFD using the idea of coefficient optimization and splitting in one-way wave-equation migration (Ma, 1983; Lee and Suh, 1985; Holberg, 1988; Soubaras, 1996). We present three different formulas for splitting by introducing auxiliary variables. Inversion of a multidiagonal matrix can be transformed into solving a series of tridiagonal matrices. Using the splitting schemes, the computation complexity of the second space derivative approximation has successfully been reduced. The stencil coefficients used in the proposed method can also be calculated with optimization in the wavenumber domain. Finally, we obtain efficient splitting high-order CFD methods with spectral-like resolution for wave-equation forward modeling.

In the remaining sections, the numerical dispersion relationships of higher order CFD schemes are first examined. Then, the splitting algorithms and their optimized implementation are proposed. Finally, numerical experiments are presented to illustrate the performance of the proposed algorithms.

METHODS

High-order CFD scheme and dispersion relation

Let us first consider an acoustic-wave equation:

$$\frac{1}{c(x,z)^2}\frac{\partial^2 P(x,z)}{\partial t^2} = \frac{\partial^2 P(x,z)}{\partial x^2} + \frac{\partial^2 P(x,z)}{\partial z^2} + s(x,z,t), \quad (1)$$

where x and z are the spatial coordinates, $c(x,z)$ is the velocity, $P(x,z)$ denotes the wavefield, and $s(x,z,t)$ denotes the source term.

Equation 1 can be solved numerically using a central FD in the time direction and a high-order CFD scheme in the space direction. The CFD scheme for approximating the second space derivatives can be derived as (Lele, 1992)

$$P_i'' + \sum_{m=1}^{M} b_m (P_{i+m}'' + P_{i-m}'') \approx \frac{1}{h^2}\left(\sum_{m=1}^{N} a_m (P_{i+m} - 2P_i + P_{i-m})\right), \quad (2)$$

where P_i and P_i'' are the wavefield and its second derivatives in grid points indexed by i, respectively, and h is the space sample step; N and M are two integers greater than zero to denote the order of the scheme. For easy analysis, we set N equal to M. The a and b are the FD coefficients. If we set all values of b to be zero, equation 2 would degenerate into a high-order EFD. Otherwise, the left side of equation 2 would be a multidiagonal matrix having $(2M+1)$ diagonals. Inverting the multidiagonal matrix with a chasing method and considering only the multiplication and division operations, the computation complexity is approximately $O((M^2+3M+1)n)$. Here, n is the number of sample grids. The right side of equation 2 is a high-order EFD and its computation complexity is approximately $O((M+1)n)$. As a result, the total computation complexity for equation 2 is approximately $O((M^2+4M+2)n)$ and most of the computation occurs when solving the multidiagonal matrix.

Coefficients a and b can be obtained using Taylor's theorem, though the resulted scheme has relatively low accuracy in high-wavenumber zones. To obtain a spectral-like resolution scheme, we will calculate coefficients a and b with optimization methods. Applying the Fourier transform to both sides of equation 2 and setting N equal to M, we obtain the dispersion relation for high-order CFD:

$$\theta^2 \approx \frac{2\sum_{m=1}^{M} a_m(1-\cos(m\theta))}{1+2\sum_{m=1}^{M} b_m \cos(m\theta)}, \quad (3)$$

where $\theta = kh$, k is the wavenumber, and $\theta = [0,\pi]$. The left side of equation 3 is the solution of the spectral method, which we use as the exact solution for comparisons. The right side is the approximated solution via the CFD. Based on least-squares (LS) optimization, the CFD coefficients a and b are optimized within the entire wavenumber interval. The fitting errors are set to be the difference between the exact solutions and the approximated solutions. In the following section, we will first illustrate our splitting algorithms for the CFD methods, and then we will discuss the computation of the coefficients using LS optimization in detail.

Splitting algorithms for the high-order CFD scheme

Transforming equation 3 back into the space domain would result in a multidiagonal matrix to invert. To overcome this problem, we use the idea of splitting in one-way wave-equation migration to propose three different ways to split equation 3. Their computation complexity is also analyzed. For clarity and simplicity, we next illustrate our method with $M=3$, being straightforward for higher M. It is straightforward for the case of M greater than three.

Algorithm 1

In equation 3, the denominator and numerator are all polynomials of degree M. The factorization of the denominator gives

$$\theta^2 \approx \frac{2\sum_{m=1}^{M} c_m(1-\cos(m\theta))}{\prod_{m=1}^{M}(f_m + 2\cos\theta)}, \quad (4)$$

where c and f are the optimized FD coefficients. For $M=3$, in the wavenumber domain, we introduce auxiliary variables:

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$$\tilde{Q}_1 = \frac{-2\sum_{m=1}^{3} c_m(1-\cos(m\theta))\tilde{P}}{h^2(f_1+2\cos\theta)},$$
$$\tilde{Q}_2 = \frac{\tilde{Q}_1}{f_2+2\cos\theta},$$
$$\tilde{Q}_3 = \frac{\tilde{Q}_2}{f_3+2\cos\theta}, \tag{5}$$

where $\tilde{P}$ is the wavefield in the wavenumber domain and $\tilde{Q}_1$, $\tilde{Q}_2$, and $\tilde{Q}_3$ are the auxiliary variables in the wavenumber domain.

Substituting equation 5 into equation 4 yields

$$k^2\tilde{P} \approx -\tilde{Q}_3. \tag{6}$$

Taking the inverse Fourier transform of equation 6, we obtain

$$P'' \approx Q_3. \tag{7}$$

Taking the inverse Fourier transform of the auxiliary variables in equation 5, we have

$$(Q_1)_{i-1} + f_1(Q_1)_i + (Q_1)_{i+1} = \frac{1}{h^2}\left(\sum_{m=1}^{3} c_m(P_{i-m} - 2P_i + P_{i+m})\right),$$
$$(Q_2)_{i-1} + f_2(Q_2)_i + (Q_2)_{i+1} = Q_1^i,$$
$$(Q_3)_{i-1} + f_3(Q_3)_i + (Q_3)_{i+1} = Q_2^i, \tag{8}$$

where the subscript denotes the grid location. Combining equations 7 and 8 allows us to solve the second space derivatives. Equation 8 consists of M tridiagonal matrices ($M = 3$). The computation complexity is approximately $O(5n)$ for solving a tridiagonal matrix, and in summary, the complexity is approximately $O((6M+1)n)$ for solving equation 4. Compared with the complexity $O((M^2+4M+2)n)$ calculated by inverting a multidiagonal matrix, the splitting algorithm successfully reduces the computation from power growth to linear growth.

Algorithm 2

Equation 3 can also be expressed as the sum of a series of short fractions

$$\theta^2 \approx \sum_{m=1}^{M} \frac{2d_m(1-\cos\theta)}{f_m+2\cos\theta}, \tag{9}$$

where d and f are the optimized FD coefficients. For $M = 3$, in the wavenumber domain, we introduce auxiliary variables:

$$\tilde{Q}_1 = \frac{-2d_1(1-\cos\theta)\tilde{P}}{h^2(f_1+2\cos\theta)},$$
$$\tilde{Q}_2 = \frac{-2d_2(1-\cos\theta)\tilde{P}}{h^2(f_2+2\cos\theta)},$$
$$\tilde{Q}_3 = \frac{-2d_3(1-\cos\theta)\tilde{P}}{h^2(f_3+2\cos\theta)}. \tag{10}$$

Substituting equation 10 into equation 9 and taking the inverse Fourier transform, we obtain

$$P'' \approx Q_1 + Q_2 + Q_3. \tag{11}$$

Taking the inverse Fourier transform of the auxiliary variables in equation 10, we have

$$(Q_1)_{i-1} + f_1(Q_1)_i + (Q_1)_{i+1} = \frac{d_1(P_{i-1} - 2P_i + P_{i+1})}{h^2},$$
$$(Q_2)_{i-1} + f_2(Q_2)_i + (Q_2)_{i+1} = \frac{d_2(P_{i-1} - 2P_i + P_{i+1})}{h^2},$$
$$(Q_3)_{i-1} + f_3(Q_3)_i + (Q_3)_{i+1} = \frac{d_3(P_{i-1} - 2P_i + P_{i+1})}{h^2}. \tag{12}$$

Combining equations 11 and 12 allows us to solve the second space derivatives.

Similar to algorithm 1, a multidiagonal matrix would still need to be split into M tridiagonal matrices, and the computation is on the order of $O((6M+1)n)$. However, unlike algorithm 1, in which the auxiliary variables are solved sequentially, here the auxiliary variables are solved in parallel.

Algorithm 3

In the following, we propose the third splitting algorithm. We modify equation 9:

$$\theta^2 \approx e_0 + \sum_{m=1}^{M} \frac{e_m}{f_m+2\cos\theta}, \tag{13}$$

where e and f are the FD coefficients. Comparing to equation 9, we have

$$e_0 = -\sum_{m=1}^{M} d_m,$$
$$e_m = d_m(2+f_m), \quad m = 1, 2, \ldots, M. \tag{14}$$

For $M = 3$, we introduce the auxiliary variables:

$$\tilde{Q}_1 = \frac{\tilde{P}}{f_1+2\cos\theta},$$
$$\tilde{Q}_2 = \frac{\tilde{P}}{f_2+2\cos\theta},$$
$$\tilde{Q}_3 = \frac{\tilde{P}}{f_3+2\cos\theta}. \tag{15}$$

Substituting equation 15 into equation 13 and taking the inverse Fourier transform, we obtain

$$P'' \approx \frac{-e_0P - e_1Q_1 - e_2Q_2 - e_3Q_3}{h^2}. \tag{16}$$

Taking the inverse Fourier transform of the auxiliary variables in equation 15, we have

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$$(Q_1)_{i-1} + f_1(Q_1)_i + (Q_1)_{i+1} = P_i,$$
$$(Q_2)_{i-1} + f_2(Q_2)_i + (Q_2)_{i+1} = P_i,$$
$$(Q_3)_{i-1} + f_3(Q_3)_i + (Q_3)_{i+1} = P_i. \quad (17)$$

Compared with algorithms 1 and 2, algorithm 3 has the same computation $O((6M+1)n)$.

Algorithm 3 has a form similar to the recursive spatial derivative operators proposed by Kosloff et al. (2008, 2010). The ideas and applicability of our algorithm are different. In this paper, the original dispersion relation formula is converted to short fraction forms according to the factorization rules. For equation 2, when M is less than N, algorithm 1 and Kosloff's recursive method can be applied. When M is greater than N, only algorithm 1 can be applied. Algorithms 2 and 3 are a special application when N is equal to M. From equations 8, 12, and 17, algorithm 3 does not need to solve the RHS EFD term and makes programming easier. Thus, modeling with algorithm 3 seems a good choice.

Table 1 shows the CPU time for solving band-limited linear equations using different methods. Each method does the calculation 2000 times and averages them to obtain an average CPU time. The time of the inverse matrix is not included in the direct inverse method. As illustrated in Table 1, the efficiency of the splitting method is increased two to three times compared with the prefactored method. Thus, the splitting method is an effective approach to solve band-limited linear equations.

Calculation of the FD coefficients via LS optimization

The FD coefficients involved in the splitting formulas can be calculated via the method of undetermined coefficients according to the original dispersion relation of equation 3. However, the method of undetermined coefficients during implementation is complex. We use an optimization method to directly calculate those coefficients.

The original nonsplitting CFD is defined in equation 3. We compute the FD coefficients by solving a linear LS optimization problem with the following misfit function:

$$E = \int_0^w \left(2\sum_{m=1}^{M} a_m(1-\cos(m\theta)) - \theta^2\left(1+2\sum_{m=1}^{M} b_m \cos(m\theta)\right)\right)^2 d\theta, \quad (18)$$

Table 1. Comparisons of the CPU time for solving a band-limited linear equations using different methods.

Matrix size	Inverse method (ms)	Prefactored method (ms)		Splitting method (ms)	
		$M=2$	$M=3$	$M=2$	$M=3$
500×500	13.43	0.34	0.38	0.10	0.16
1000×1000	57.88	0.63	0.70	0.19	0.27
1500×1500	138.37	0.93	1.02	0.26	0.40
2000×2000	255.80	1.19	1.32	0.34	0.55
2500×2500	389.92	1.47	1.65	0.45	0.64

where $0 < w \le \pi$. The LS solution of minimizing the misfit function 18 exists and is globally optimal. Solving this equation has been discussed in great detail (Kosloff, 2010; Zhou and Zhang, 2011; Liu, 2013). Here, we focus on the calculation of the splitting CFD coefficients via LS optimization. After obtaining the optimized coefficients a and b in equation 3, we define the following misfit functions for calculating the coefficients in our splitting algorithms:

$$E=\int_0^w \left(\frac{2\sum_{m=1}^{M} c_m(1-\cos(m\theta))}{\prod_{m=1}^{M}(f_m+2\cos\theta)} - \frac{2\sum_{m=1}^{M} a_m(1-\cos(m\theta))}{1+2\sum_{m=1}^{M} b_m\cos(m\theta)}\right)^2 d\theta. \quad (19)$$

$$E=\int_0^w \left(\sum_{m=1}^{M}\frac{2d_m(1-\cos\theta)}{f_m+2\cos\theta} - \frac{2\sum_{m=1}^{M} a_m(1-\cos(m\theta))}{1+2\sum_{m=1}^{M} b_m\cos(m\theta)}\right)^2 d\theta. \quad (20)$$

$$\varepsilon_{\mathrm{rms}}=\sqrt{\frac{1}{w}\int_0^w \left(2\sum_{m=1}^{M} a_m(1-\cos(m\theta)) - \theta^2\left(1+2\sum_{m=1}^{M} b_m\cos(m\theta)\right)\right)^2 d\theta}. \quad (21)$$

The misfit functions defined in equations 19 and 20 are for algorithms 1 and 2, respectively.

Minimizing the errors of equations 19 and 20 are nonlinear LS problems, which can be solved iteratively with line search methods or trust region methods. Because they are robust to ill-conditioned problems (Yuan, 1999), we selected trust region methods to solve the nonlinear LS optimization problems 19 and 20. During implementation, we use the function *lsqcurvefit* in the MATLAB® Toolbox for the optimization of the objective functions. Only the misfit function value in equation 19 is needed as input, and no other values, such as the Jacobi of the misfit function, are needed because the *lsqcurvefit* will calculate the Jacobi misfit function value by numerical methods. For the numerical integration over θ, we sample 100 grid points and we choose random numbers of $[-10, 10]$ as the initial values for c and f. We initially observed that only several iterations would guarantee convergences. The solution of equation 19 will generate the optimized coefficients c and f that have been defined in equation 4. In addition, we assume that the coefficient f for these equations is the same and, therefore, is available for equation 20, too, which can be used to obtain the optimized coefficient d in equation 20. After d and f are determined, the coefficient e used in algorithm 3 of equation 13 can be calculated by equation 14.

Equation 21 defined the rms error to examine the accuracy. Table 2 lists the optimized coefficients for the nonsplitting and splitting algorithms. The optimization intervals are $w = 1.73$, 2.51, and 2.98 for $M = 1$, 2, and 3, respectively.

Figure 1 shows the comparison of the variations in the errors with θ, for the optimized CFD and optimized EFD methods. For $M = 1$, the optimized CFD can achieve the same accuracy for $M = 4$ by the EFD method. The accuracy of the CFD with $M = 3$ is much greater than the LSEFD with $M = 8$.

Figure 2 shows variations in the errors with θ for the splitting and nonsplitting algorithms. From the curves, the splitting algorithms preserve the same accuracy as the nonsplitting algorithm, and all three algorithms have the same level of accuracy.

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From Figure 2, we draw a rough deduction that the stability of our splitting algorithms should be similar to the nonsplitting CFD method. However, guaranteeing the accuracy near the model boundary remains a problem to be solved in the future.

NUMERICAL EXAMPLES

In this section, we will compare the effectiveness of our proposed splitting-based CFD method with conventional FDMs. We first evaluate them in a constant media experiment. The constant velocity is 2000 m/s, and the horizontal and vertical spacings are 20 m. The extrapolation time step is 2 ms. The simulation time is approximately 0.9 s. The Ricker wavelet with a peak frequency of 20 Hz is used as the source. We simulate the wavefield using the CFD method with the coefficients calculated using the following methods: the Taylor theory CFD for $M = 1$ (Figure 3a), the optimized CFD for $M = 1$ (Figure 3b), the nonsplitting optimized CFD for $M = 2$ (Figure 3c), the splitting optimized CFD algorithm 1 for $M = 2$ (Figure 3d), the splitting optimized CFD algorithm 2 for $M = 2$ (Figure 3e), the splitting optimized CFD algorithm 3 for $M = 2$ (Figure 3f), the splitting optimized CFD algorithm 3 for $M = 3$ (Figure 3g), the optimized EFD for $M = 4$, with the coefficients $c_0 = -2.968111$, $c_1 = 1.700010$, $c_2 = -0.2554615$, $c_3 = 0.04445392$, and $c_4 = -0.004946851$ from Liu (2013), in Table 3 (Figure 3h), and the spectral method (Figure 3i). Figure 3 illustrates a snapshot of the wavefield at $t = 0.9$ s. Figure 4 shows the selected trace at sample ix = 101 in Figure 3. The optimized CFD method results in higher accuracy (comparing trace 1 with trace 2 in Figure 4), and all of the methods have correct results. Specifically, the splitting and nonsplitting-based CFD methods yield the same level of accuracy (comparing the nonsplitting result of trace 3 with the splitting result of traces 4, 5, and 6).

The second example is the numerical modeling of the wave propagation in a 2D Sigsbee model by The Subsalt Multiples Attenuation and Reduction Technology Joint Venture. The velocity model is shown in Figure 5. For the computation, we set 503 points in the horizontal direction and 254 points in the vertical direction. The horizontal and vertical grid sizes are 20 m. The total simulation time is 6 s. The source is a Ricker wavelet with a peak frequency of 20 Hz at the position of $z = 1000$ m and $x = 5020$ m (denoted as

Table 2. Optimized FD coefficients for the nonsplitting and splitting algorithms.

Coefficients	$M = 1$	$M = 2$	$M = 3$
a_1	1.2344794295	0.4608059195	−0.0637630527
a_2	—	0.3548546586	0.4271156503
a_3	—	—	0.0993539529
b_1	0.1263979096	0.4164568096	0.6251536295
b_2	—	0.0256498598	0.1393910977
b_3	—	—	0.0065278704
c_1	9.7666127010	17.9652412687	−9.7678155056
c_2	—	13.8345652407	65.4295770709
c_3	—	—	15.2199661501
d_1	9.7666127010	13.1168703858	1.0386408854
d_2	—	0.7176948549	0.0625873459
d_3	—	—	14.1187379186
e_0	−9.7666127010	−13.8345652407	−15.2199661499
e_1	96.8020088015	203.2534914832	5.4377750680
e_2	—	3.4023354003	0.25890804182
e_3	—	—	253.8691915385
f_1	7.9115232441	13.4955782519	3.2354718019
f_2	—	2.7406434323	2.1367474248
f_3	—	—	15.9810116884

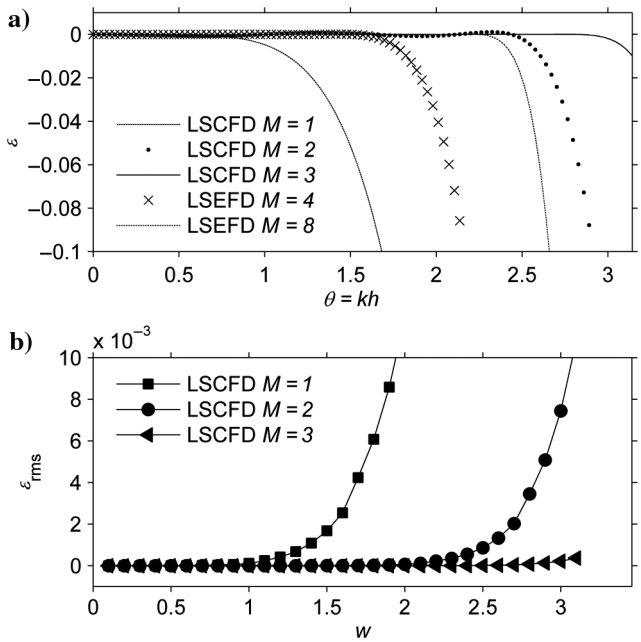

Figure 1. (a) Comparisons of the variations in the error with θ for the optimized CFD and EFD. (b) Variations in the rms error with w for different values of M.

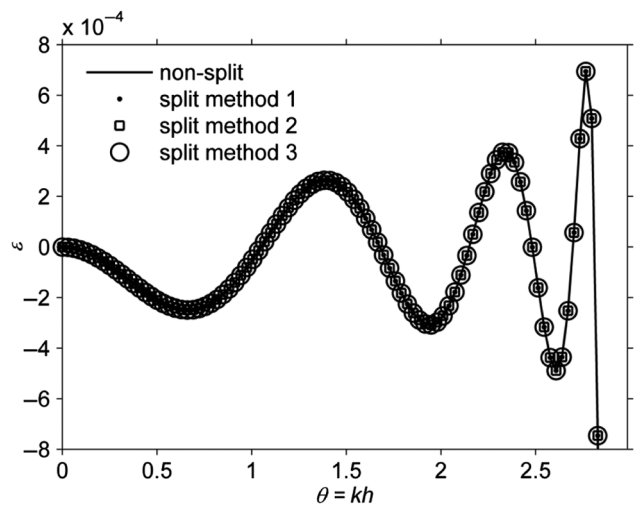

Figure 2. Variations in the error ε with θ for the splitting and nonsplitting methods with $M = 3$.

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"+" in Figure 5). The PML absorbing boundary conditions are used at all boundaries, and the absorbing boundary layers have a thickness of 10 grids. Figure 6 shows a snapshot of the wavefield at $t = 1.5$ s, calculated by the eighth-order optimal EFD and the high-order CFD with $M = 3$. In Figure 6a, the wavefront calculated by the optimal EFD has an obvious numerical dispersion in the low-velocity zones, whereas the numerical dispersion is greatly reduced in Figure 6b. This difference is mainly due to the spectral resolution of the high-order CFD scheme. Figure 7 shows the shot records. The depth of the receivers is 1000 m.

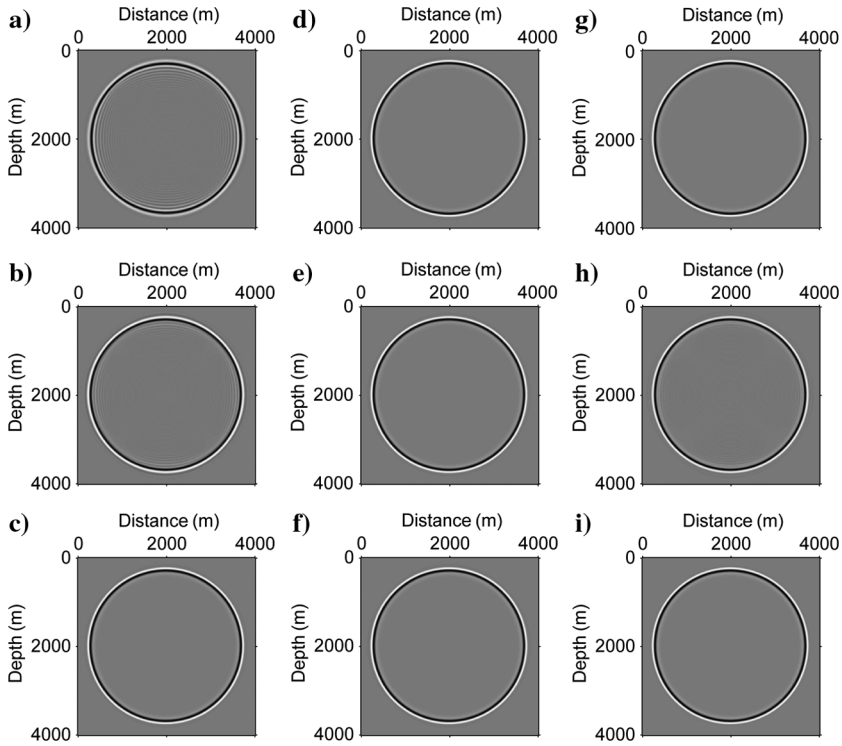

Figure 3. Snapshots for (a) $M = 1$, CFD method by Taylor; (b) $M = 1$, optimized CFD; (c) $M = 2$, nonsplitting CFD; (d) $M = 2$, splitting optimized CFD, algorithm 1; (e) $M = 2$, splitting optimized CFD, algorithm 2; (f) $M = 2$, splitting optimized CFD, algorithm 3; (g) $M = 3$, splitting optimized CFD, algorithm 3; (h) $M = 4$, optimized EFD; and (i) spectral method.

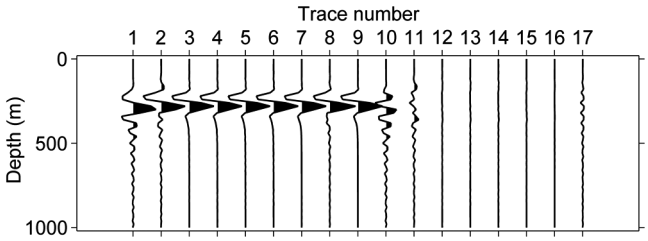

Figure 4. Comparisons of the selected traces in Figure 3. Traces 1–9 with x-coordinate of 2000 m are shown in Figure 3a–3i, respectively. Traces 10–17 are the differences between 1–8 and 9, respectively.

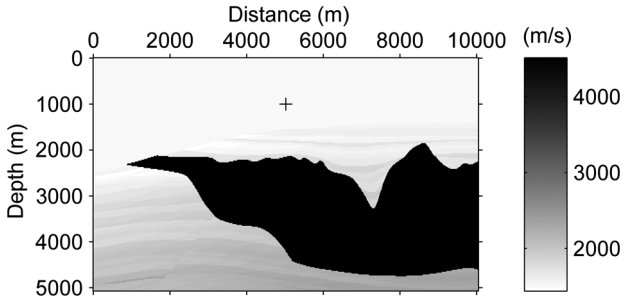

Figure 5. Velocity model of the 2D Sigsbee model. The "+" denotes the source position.

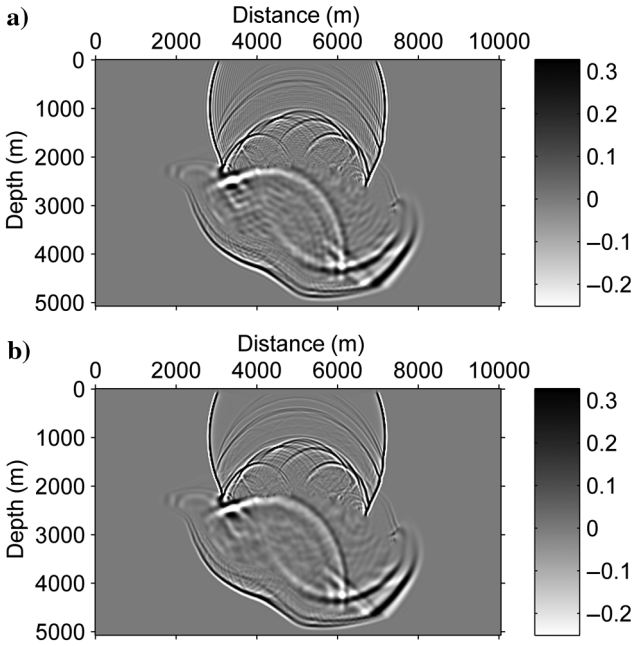

Figure 6. Snapshot at $t = 1.5$ s for the 2D model shown in Figure 5 using (a) the eighth-order optimal EFD and by (b) the CFD method with $M = 3$, respectively.

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DOI:10.1190/geo2015-0418.1

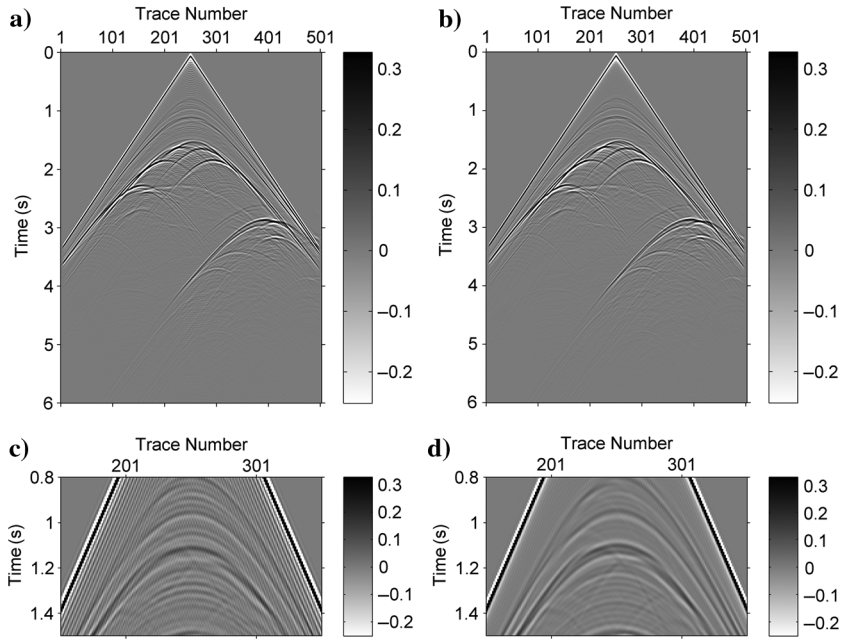

Figure 7. Shot records for the 2D model shown in Figure 3 by (a) the eighth-order optimal EFD and by (b) the CFD method with $M = 3$, respectively. Panels (c and d) are the magnified results of (a and b), respectively.

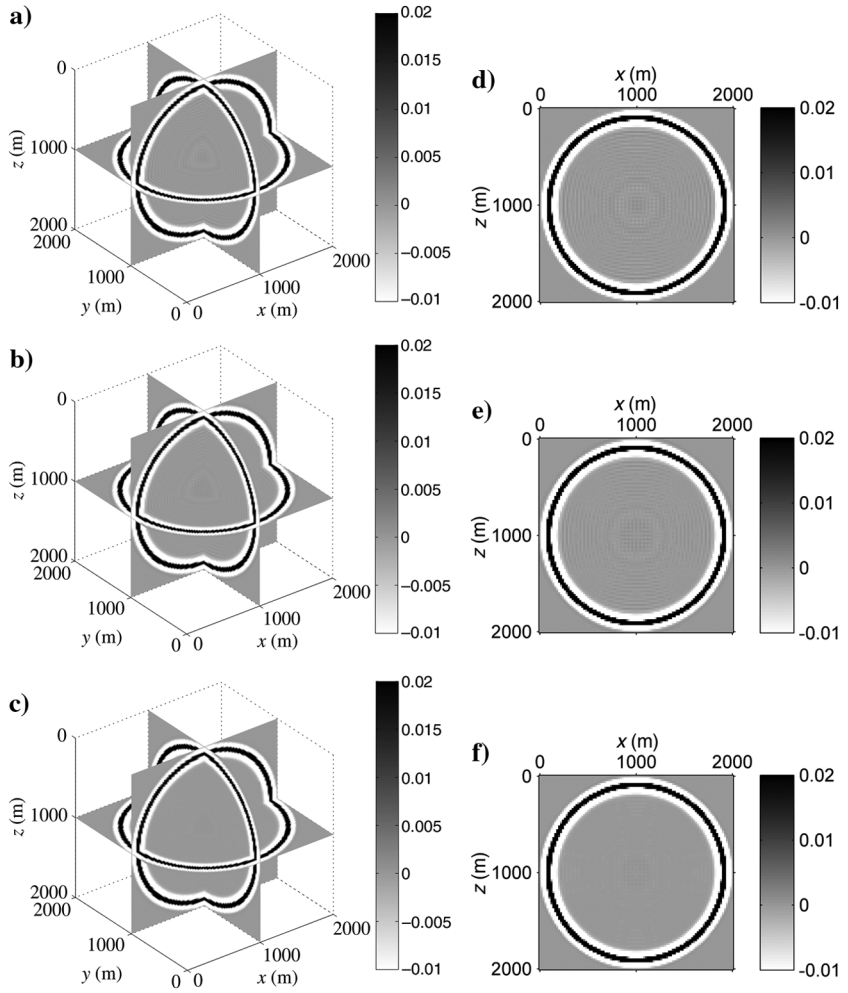

Figure 8. Snapshot slices of the 3D test, with snapshot slices at $t = 0.5$ s for (a) LSEFD $M = 8$, (b) CFD $M = 2$, (c) CFD $M = 3$ and vertical slice with $y =$ coordinate of 1000 m by (d) LSEFD $M = 8$, (e) CFD $M = 2$, and (f) CFD $M = 3$, respectively.

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The third example we present is a 3D model with a constant velocity of 2000 m/s. The grid points of all three directions are 101 with a step of 20 m. The calculation time is up to 0.5 s, with a time step of 2 ms. The 20 Hz Ricker wavelet is placed at a selected grid position ($z = 1000$ m, $x = 1000$ m, $y = 1000$ m). In Figure 8, the LSEFD with $M = 8$ (Figure 8a and 8d) and CFD with $M = 2$ (Figure 8b and 8e) give results of comparable accuracy. The modeling accuracy of the CFD with $M = 3$ (Figure 8c and 8f) is much greater than the LSEFD with $M = 8$ (Figure 8a and 8d). Runtimes of the LSEFD and CFD with $M = 2$, 3 are 496, 350, and 526 s, respectively.

CONCLUSIONS

In this paper, efficient splitting algorithms for high-order CFD are proposed. The algorithms transform the inversion of the multidiagonal matrix in the high-order CFD into an inversion of a series of tridiagonal matrices. Three different splitting algorithms are proposed, and numerical examples verified the correctness and effectiveness of our algorithms. The experiments verified that we achieve considerably faster computation times with little loss in accuracy. The corresponding computation analysis shows that the computation burden can be reduced from exponential growth to linear growth. This reduction makes the method more feasible for practical applications.

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